Metasurface Geometric Polarization Matrix-Tensor Computing Accelerator: O(1) Unitary Matrix Inversion for AGI-Scale Linear Algebra Offloading

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Abstract

We propose a dedicated Metasurface Geometric Polarization Matrix-Tensor Computing Accelerator (MGP-MTCA), a lightweight extended architecture of B^3D-HPV focused on pure metasurface-based photonic acceleration. By engineering subwavelength metasurfaces to manipulate the intrinsic polarization state of incoherent light via geometric modulation, this accelerator enables deterministic execution of high-dimensional unitary matrix multiplication, tensor operation and O(1) complexity unitary matrix inversion at the speed of light with near-zero static power consumption. Distinguished from phase-dependent interferometric photonic computing, the MGP-MTCA leverages the Rigid Geometry of Polarization (RGoP), endowing the system with ultrahigh thermal robustness and complete immunity to phase drift and coherent crosstalk. A simplified software-programmable electrical interface enables direct, low-latency integration with silicon-based GPUs/TPUs, achieving efficient offloading of core linear algebra tasks (matrix/tensor operations) from silicon computing cores. This pure metasurface architecture abandons complex optical detection and 3D lattice substrates, providing a highly industrializable, low-entropy photonic acceleration solution for AGI-scale matrix-tensor computing.

I. Introduction: Silicon Linear Algebra Bottleneck and Geometric Polarization Computing Solution

Current AGI-scale computing is severely constrained by the intrinsic inefficiency of silicon-based architectures in executing high-dimensional matrix-tensor operations and matrix inversion—core linear algebra tasks that dominate neural network training/inference, inverse problem solving and tensor decomposition. Silicon multiply-accumulate (MAC) units rely on iterative digital logic and charge transfer, leading to high entropy generation, limited parallelism and fundamental latency bottlenecks, with matrix inversion exhibiting polynomial time complexity that exacerbates the "Power Wall" and "Memory Wall".

To address this bottleneck, we propose a pure metasurface geometric polarization computing paradigm, which shifts the burden of matrix-tensor computing from silicon iterative operations to physical optical geometric evolution. By mapping high-dimensional matrix-tensor data to the Poincaré Sphere and encoding linear algebra operations as geometric polarization transformations on subwavelength metasurfaces, mathematical computations are no longer "solved" via digital iteration, but "revealed" through the intrinsic symmetry and propagation evolution of the polarization field. This paradigm abandons complex 3D optical substrates and multi-channel photodetection, focusing on

the minimal design of metasurface-only geometric polarization matrix-tensor computing, enabling immediate industrialization and seamless integration with existing silicon computing ecosystems.

II. Core Principle: Geometric Polarization Inversion (GPI) for O(1) Matrix Inversion and Matrix-Tensor Operation

The MGP-MTCA redefines core linear algebra operations—unitary matrix multiplication, high-dimensional tensor operation and unitary matrix inversion—as intrinsic geometric polarization transformations on metasurfaces, with matrix inversion realized via the Geometric Polarization Inversion (GPI) principle to achieve O(1) time complexity, the theoretical lower bound of computing latency.

2.1 Geometric Polarization Encoding of Matrix-Tensor Data

High-dimensional unitary matrices and tensors are directly encoded as spatial geometric polarization transformation profiles on the metasurface. Each element of the matrix/tensor is mapped to a specific polarization state (linear, circular, elliptical) and its geometric modulation parameters on the metasurface, with the matrix-tensor product and linear combination equivalent to the superposition and cascade of geometric polarization transformations on the metasurface. This encoding method realizes the direct physical mapping of matrix-tensor data, where the metasurface itself serves as the computing substrate—"the metasurface is the matrix, the polarization evolution is the tensor operation".

2.2 O(1) Complexity Unitary Matrix Inversion via GPI

For any unitary transformation $\boldsymbol{M}$ encoded by the metasurface via geometric polarization modulation, the matrix inverse $\boldsymbol{M}^{-1}$ is physically and uniquely equivalent to the Hermitian Adjoint $\boldsymbol{M}^\dagger$ of the transformation matrix, a fundamental property derived from the unitary invariance of geometric polarization transformations on the Poincaré Sphere.

Matrix inversion is realized through two intrinsic physical properties of the metasurface: (1) geometric reconfigurability—reversing the spatial geometric orientation of the metasurface's subwavelength meta-atoms to directly generate the Hermitian Adjoint transformation profile; (2) polarization reciprocity—inverting the propagation direction of the input polarization field through the fixed metasurface to achieve the polarization geometric transformation corresponding to $\boldsymbol{M}^\dagger$. In both implementations, matrix inversion occurs as a spontaneous physical collapse of the polarization state, with computation latency only determined by the light propagation time through the metasurface, independent of the matrix/tensor dimension—realizing strict O(1) time complexity for unitary matrix inversion.

2.3 Intrinsic Parallelism of Geometric Polarization Tensor Operation

High-dimensional tensor operations are realized via the spatial and polarization domain parallelism of the metasurface. Orthogonal polarization states (s-polarization, p-polarization, left/right circular polarization) on the metasurface serve as independent, non-interfering tensor computing channels, with each channel executing parallel tensor sub-operation via geometric polarization modulation. The superposition of orthogonal polarization channels on the metasurface realizes the full-dimensional tensor operation, with parallelism scaling linearly with the number of orthogonal polarization states—achieving massive parallelism for AGI-scale tensor computing without coherent crosstalk.

III. Pure Metasurface Realization: Material and Geometric Modulation Design

The MGP-MTCA adopts a pure metasurface architecture with no additional 3D optical substrates, complex photodetection modules or multi-channel optical coupling components. The core computing substrate is a single-layer planar metasurface engineered from CMOS-compatible dielectric materials, with all matrix-tensor operations and O(1) matrix inversion realized via the geometric modulation of polarization states by the metasurface's subwavelength meta-atoms.

3.1 Metasurface Material Selection

The metasurface is fabricated from high-refractive-index, low-loss CMOS-compatible dielectric materials (e.g., silicon, silicon nitride, titanium dioxide), which exhibit excellent birefringence and geometric polarization modulation capabilities in the near-infrared communication band (1310 nm/1550 nm). The dielectric material system ensures high thermal stability, low optical loss and seamless integration with existing silicon semiconductor fabrication processes, enabling large-scale batch manufacturing and low-cost industrialization. The subwavelength meta-atoms are patterned into periodic/aperiodic pillar, slit or antenna structures via standard photolithography and etching processes, with the geometric parameters (size, aspect ratio, spatial orientation) of each meta-atom precisely tailored to realize deterministic geometric polarization modulation.

3.2 Binary-Driven Geometric Polarization Modulation

Each subwavelength meta-atom on the metasurface acts as a fundamental geometric polarization computing unit, controlled by a simple CMOS-compatible binary electrical signal (0/1). The binary signal modulates the local birefringence state of the meta-atom by tuning its geometric or optical properties, thereby switching the geometric polarization transformation parameter encoded by the meta-atom. This binary-driven modulation scheme enables fast reconfiguration of the metasurface's matrix-tensor encoding profile, with reconfiguration speed matching the real-time computing requirements of

silicon-based GPUs/TPUs. The binary control interface is designed for direct electrical integration with silicon computing cores, eliminating complex optical-electrical conversion circuits and reducing system integration complexity.

3.3 Rigid Geometry of Polarization (RGoP) for Thermal Robustness

The MGP-MTCA leverages the Rigid Geometry of Polarization (RGoP)—the intrinsic property that geometric polarization transformations on the metasurface are determined by the fixed spatial geometry of meta-atoms, independent of phase and environmental temperature fluctuations. This property endows the pure metasurface architecture with ultrahigh thermal robustness, with no performance degradation under typical industrial temperature variations (-40 °C to 85 °C). Unlike phase-dependent interferometric photonic computing, the MGP-MTCA requires no phase calibration, thermal compensation or active feedback control, realizing passive, stable matrix-tensor computing and O(1) matrix inversion—greatly simplifying the system design and improving long-term operational reliability.

IV. Simplified Silicon-Metasurface Integration Interface

To enable efficient offloading of matrix-tensor tasks from silicon computing cores, the MGP-MTCA adopts a simplified software-programmable electrical interface for direct low-latency integration with silicon-based GPUs/TPUs. This interface abandons complex multi-channel photodetection and Stokes parameter reconstruction, realizing the direct conversion of the metasurface's geometric polarization computing results to digital electrical signals that can be directly consumed by silicon computing cores.

The input side of the interface receives digital matrix-tensor data from the GPU/TPU and converts it into binary control signals for the metasurface, programming the geometric polarization transformation profile of the metasurface in real time. The output side captures the integrated optical signal of the polarization state after geometric evolution on the metasurface and converts it into a low-jitter digital electrical signal via a simple high-speed photodetector and comparator, with a software-programmable reference voltage enabling dynamic adjustment of computing precision to match different AGI-scale linear algebra tasks.

This simplified interface enables the MGP-MTCA to offload the majority of core linear algebra tasks (matrix multiplication, tensor operation, unitary matrix inversion) from silicon computing cores, with the GPU/TPU only retaining control logic, non-linear operation and post-processing tasks. The pure metasurface accelerator executes all high-dimensional matrix-tensor operations at the speed of light with near-zero power consumption, significantly reducing the power consumption and computational load of silicon-based computing systems while improving the overall computing efficiency of AGI-scale systems.

V. Computational Invariance and Passive Robustness

The MGP-MTCA inherits the Logical Topological Equivalence Principle for geometric polarization computing: as long as the global geometric polarization transformation f(x) (i.e., the unitary matrix/tensor encoded by the metasurface) is preserved, the specific propagation routing of the light field through individual meta-atoms is irrelevant. This principle endows the pure metasurface architecture with intrinsic passive robustness and fault tolerance, without the need for active error correction or hardware redundancy.

For defective or degraded meta-atoms on the metasurface, the global geometric polarization transformation f(x) remains unchanged due to the topological invariance of the metasurface's polarization encoding profile, ensuring the correctness of matrix-tensor computing and O(1) matrix inversion. This passive fault tolerance enables the MGP-MTCA to operate stably with a high meta-atom yield threshold, greatly reducing the manufacturing precision requirements and industrialization costs of the metasurface.

VI. Conclusion

The Metasurface Geometric Polarization Matrix-Tensor Computing Accelerator (MGP-MTCA) is a highly industrializable pure metasurface photonic accelerator dedicated to AGI-scale linear algebra offloading, focusing on metasurface-only geometric polarization matrix multiplication, tensor operation and O(1) unitary matrix inversion. By abandoning complex 3D optical substrates and multi-channel photodetection modules, this accelerator leverages the Rigid Geometry of Polarization (RGoP) on CMOS-compatible dielectric metasurfaces to realize physical geometric evolution of matrix-tensor data, with computation at the speed of light, near-zero power consumption and strict O(1) time complexity for unitary matrix inversion.

The pure metasurface architecture enables seamless integration with existing silicon semiconductor fabrication processes and silicon-based GPUs/TPUs, providing a low-cost, high-reliability photonic acceleration solution for the silicon linear algebra bottleneck. As a lightweight extended architecture of B^3D-HPV, the MGP-MTCA verifies the feasibility of geometric polarization computing on pure metasurfaces, laying a solid foundation for the development of next-generation photonic computing accelerators for AGI-scale matrix-tensor computing.

This accelerator redefines the role of metasurfaces in photonic computing: from passive optical components to active matrix-tensor computing substrates. By adhering to the design philosophy of "computing as physical geometric evolution", the MGP-MTCA opens a new path for low-entropy, sustainable photonic computing in the post-Moore era.

Appendix A: Mathematical Formulation of Incoherent Polarization Computing

A.1 Data Mapping via Stokes Formalism

Unlike phase-based computing which uses complex amplitudes (E-field), MGP-MTCA operates on the Incoherent Power Domain using the Stokes Vector $\vec{S}$:

$$\vec{S} = \begin{pmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{pmatrix} = \begin{pmatrix} I \\ I_h - I_v \\ I_{45} - I_{-45} \\ I_r - I_l \end{pmatrix}$$

Where S_0 is total intensity, and S_1, S_2, S_3 represent the geometric coordinates on the Poincaré Sphere. A high-dimensional tensor element is mapped to a specific (θ, ϕ) on the sphere, establishing a direct one-to-one correspondence between numerical tensor data and physical polarization geometric states.

A.2 Matrix-Tensor Operation as Mueller Transformation

The interaction between the input light and the metasurface meta-atoms is described by a Mueller Matrix $\boldsymbol{M}$, the core mathematical representation of geometric polarization modulation on the metasurface:

$$\vec{S}_{out} = \boldsymbol{M}_{metasurface} \cdot \vec{S}_{in}$$

For a unitary matrix operation, the metasurface is engineered such that $\boldsymbol{M}$ is a rotation matrix in the S_1, S_2, S_3 subspace, ensuring the unitary invariance of polarization geometric transformations and the conservation of incoherent light power. Due to the linearity of the Stokes space, a 2D metasurface array executing spatial-multiplexing of independent Mueller Matrix profiles effectively performs the core linear algebra operation for AGI-scale computing:

$$\mathbf{Y} = \mathbf{W} \mathbf{X} + \mathbf{b}$$

where $\mathbf{W}$ is the spatially-variant Mueller profile encoded on the metasurface, $\mathbf{X}$ is the input high-dimensional tensor mapped to the Stokes Vector set, and $\mathbf{b}$ is the bias term implemented via global polarization state offset.

A.3 The GPI Inversion Formula

The core breakthrough of MGP-MTCA is the physical realization of the matrix inverse operation via geometric polarization modulation, eliminating the polynomial time complexity of silicon-based iterative inversion algorithms. For a unitary geometric

transformation $\boldsymbol{M}$ encoded by the metasurface, the inversion is simplified to the Hermitian Adjoint in the polarization manifold:

$$\boldsymbol{M}^{-1} \equiv \boldsymbol{M}^\dagger \approx \boldsymbol{M}^T \ (\text{for pure geometric rotation})$$

In the MGP-MTCA hardware, this mathematical inversion is realized through two physical implementations with strict O(1) time complexity:

1. Geometric Transpose: Direct reconfiguration of the metasurface meta-atom array to realize the matrix transpose $\boldsymbol{M}_{ij} \to \boldsymbol{M}_{ji}$, where the reconfiguration process is synchronized with silicon computing core control signals and independent of matrix/tensor dimension.
2. Reciprocity Projection: Utilization of the polarization reciprocity of the metasurface to achieve the inverse transformation via $\vec{S}_{out} \cdot \boldsymbol{M}^T = \vec{S}_{in}$. The computation complexity is strictly O(1) because the "calculation" is a singular pass of photons through the $\boldsymbol{M}^T$ configured metasurface, with latency only determined by the light propagation time in the dielectric medium.

Engineering Note (Core Design Hook)

"The metasurface does not simulate the matrix; it IS the matrix. The light does not solve the tensor equation; it evolves through it—geometric polarization is the language of metasurface matrix-tensor computing."

Keywords

Metasurface Photonic Accelerator; Geometric Polarization Computing; Matrix-Tensor Operation; O(1) Matrix Inversion; AGI-Scale Linear Algebra Offloading; CMOS-Compatible Dielectric Metasurface; Rigid Geometry of Polarization; Stokes Formalism; Mueller Transformation